

Part 6: Display a Message

1. **Which DOS interrupt is used to display a character?** `INT 21h` is the interrupt. Specifically, you use Function `02h`.
2. **How do you display a string using INT 21h?** Set `AH = 09h`, load the address of the string into `DX` (`LEA DX, msg`), and call `INT 21h`.
3. **Which register stores the ASCII code for a character?** For printing a single character (Function `02h`), the ASCII code goes in the **DL** register.
4. **How do you display a newline in DOS?** You must print two characters: Carriage Return (ASCII 13) and Line Feed (ASCII 10).
5. **Which function of INT 21h is used for printing a string?** Function `09h` (Put `09h` in `AH`).
6. **How do you terminate a program after displaying a message?** Use `INT 21h` with Function `4Ch`. (`MOV AH, 4Ch` then `INT 21h`).
7. **How do you store a string in memory?** Define it in the data segment using `DB` (Define Byte). Example: `msg DB 'Hello$'`.
8. **How do you print a message character by character?** Use a loop that loads a character into `DL`, calls Function `02h` (`INT 21h`), and repeats until the string ends.
9. **Can you use loops to display a string?** Yes. You can loop through the string's memory address, printing one character at a time.
10. **How do you handle \$ in string printing with DOS?** For Function `09h`, the `$` symbol marks the **end** of the string. It is not printed.
11. **How do you display numeric values as characters?** You must convert the number to ASCII by adding `30h` (for digits 0-9) before printing.
12. **Which register holds the segment of a string for printing?** The **DS** (Data Segment) register must point to the segment where the string is stored.
13. **How do you print multiple messages sequentially?** Print the first message (Load `DX`, `INT 21h`), then load the address of the second message into `DX` and call `INT 21h` again.
14. **How can you display a message in uppercase only?** You must convert the characters in memory (subtract `20h` from lowercase ASCII) before calling the print interrupt.
15. **How is a message terminated in memory for DOS printing?** It must end with the `$` character if you are using Function `09h`.